

to the task due to its intrinsic slow execution time. Electronic-system-level (ESL) simulation is of great help in starting embedded software validation before silicon but it does not offer the hardware design accuracy to trace hardware bugs," says Moorthy. Neither can you test integration of the hardware with software. This is where hardware-assisted verification comes into play.

"Platforms like Zebu map hardware part of the design into dedicated and reprogrammable hardware resources and store the software portion into local memories. It acts as an early prototype of the complete SoC design and offers a powerful debugging means into the hardware and software parts to help hardware and software teams to quickly zero-in and pinpoint design bugs," shares Moorthy.

Such tools can help you start RTL verification at the block level and move on to the system level as the entire design solidifies. These can also co-verify interaction of hardware and software once the full RTL design is complete.

Transaction-based verification

SoC design teams have started adopting high-level design languages like System C and Verilog and transaction-level modeling for test benches to speed up functional verification. It is essential that the test bench and hardware emulator communicate in real time with maximum efficiency. Transaction-based modeling helps in achieving this.

According to Moorthy, there are three specific challenges to the creation of a successful transaction-based emulation environment: transactor creation, test environment creation (or stimulus generation) and performance. "Creating a transactor, especially for a complex protocol, can be a difficult task. Some tools provide large catalogues of transactors including most popular protocols (PCIe, USB, Ethernet, AXI and Video)," he adds. You can also create custom transactors using a

Co-design & Co-verification Tool Providers

- Advanced Micro Tools
- Cadence Design Systems
- Coventor (Indian distributor-FTD Infocom)
- EVE (Indian distributor-Oxys Technologies)
- GAO Technologies
- Mentor Graphics
- Simpod
- Synopsys

SystemVerilog behavioural compiler such as ZEMI3.

Transactors connect the emulated design to the test environment and the test environment generates stimulus for the design. ZeBu transactors are easily integrated with hardware verification and software development environments in a variety of ways, making it easy to reuse your existing test environment or quickly create a new one. For hardware verification environments, transactors can be integrated with existing high-level verification language (HVL) and C/C++ test benches via the SystemVerilog direct programming interface (SV DPI) or SystemVerilog classes. For software development environments, transactors can be integrated with C, C++ and SystemC ESL environments either via DPI, API or TLM-2.0 transactor adaptor.

Another tool from Mentor Graphics, TestBench Xpress (TBX), supports SV DPI to provide building blocks that can create transaction-based communication link between HVL and HDL. The basic transaction constructs in TBX are exported and imported constructs/tasks from SV DPI. TBX automatically generates a direct communication interface between the C, C++ or System C environment on a host and the SoC device-under-test (DUT) in the Veloce emulator.

ESL tools like QEMU can also be used to provide a virtual version of the ICE environment, providing real-world stimulus to the design. Cadence Incisive Palladium series leverages advanced RTL and ESL verification automation features, such as assertion-based acceleration and

transaction-based acceleration. Arya cites, "In such tools, you can provide real-world stimuli by external equipment."

Parametric hardware blocks for modeling

Going back to the MEMS example, one approach for co-design is to do finite element modeling. Raman explains, "You can extract the so-called reduced-order models that can extract accurate MEMS behaviour with finite element modeling. These models can be implemented in simulation languages that are used in IC simulators, such as Verilog-A."

The limitation of this approach is that the design is not parametric, so in a way it doesn't support 100 per cent co-design. Raman says, "Another approach is to include a library of parametric MEMS building blocks, like the ones available in Coventor MEMS+. Most types of MEMS can be modelled by combining these blocks, which can be directly simulated using IC simulators such as Cadence Spectre or MATLAB Simulink. MEMS+ produces models that can be automatically imported to the Cadence Virtuoso and MATLAB Simulink environments accurately."

To conclude

The ultimate goal of any designer is to meet the initial requirement within the given time-frame. Adoption of development methodologies like concurrent development helps in reaching this goal. Hardware-software co-design and co-verification is a great way to augment the independent development process, thereby increasing the design productivity. The methodology of hardware modeling has advanced to a stage where we can sense the developed software on the models of hardware, making co-simulation part of a new paradigm. The result is reduced debug cycles and accelerated time-to-market. ●

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